Southeast University

Review: Introduction to Software Engineering

wanglulu,lliao@seu.edu.cn

试卷结构

1.单项选择题	1'×10	10%
2.多项选择题	2'×5	10%
3.简答题	5'×6	30%
4.问答题	10‘×2	20%
5.分析题	15'×2	30%

Unit 0 Overview

- 0. History of computer technology
  - The raise of Software Engineering
- 1. The Nature of Software
  - What is Software?
  - The nature of software
    - Developed / Deteriorates / Custom built/ Complex
  - Legacy Software &Open Source software
  - The development of our native software

Unit 0 Overview

2. Concepts of SE

- Software crisis
- Software Process
- Why we need software processes?
- Software Lifecycle
- Umbrella / Framework Activities

Unit 1 Software Processes

- 3. Software Process Structure
  - A Generic Process Model
  - CMMI
- 4. Process Models
  - Prescriptive Process Model
  - The Waterfall Model & the V Model
  - The Incremental Model
  - Evolutionary Models
    - Prototyping
    - The Spiral
  - Concurrent
  - The Unified Process (UP)

Unit 1 Software Processes

- **5. Agile Development**
  - **Manifesto for Agile Software Development**
  - **Agility and the Cost of Change**
  - **An Agile Process**
  - **Agility Principles**
  - **Extreme Programming (XP) / Scrum / Kanban**
- **6. Human Aspects of Software Engineering**

Unit 2 Modeling

- **7. Understanding Requirements**
  - System Engineering
  - Requirements Engineering
    - Inception / Elicitation / Elaboration /Negotiation Specification / Validation / Requirements management
    - Non-Functional Requirements
    - Diagrams (Use-case / Class / State / Activity)
- **8-10. Requirements Modeling**
  - Domain Analysis
  - Scenario-Based Modeling
  - Class-Based Modeling
    - The definition of classes
      - Nouns&verbs / Potencial classes / Analysis Classes
      - Attributes&Operations / CRC card
  - Behavioral Modeling

Unit 2 Modeling

- **11. Design Concepts**
  - Design and quality
  - Fundamental Concepts
    - Abstraction / Architecture / Patterns / Separation of concerns / Modularity / Hiding / Functional independence / Refinement / Aspects / Refactoring
    - OO design concepts:
      Design classes / Inheritance / Messages / Polymorphism
    - Design Classes
- **12-14.Design Methods**
  - Architecture Design
    - What is Architecture
    - The importance of architecture
    - Architectural Styles
    - Architectural Design
    - Agility and Architecture

Unit 2 Modeling

12-14.Design Methods

- Component Level Design
  - What is a Component
  - Basic Design Principles
    - The Open-Closed Principle (OCP)
    - The Liskov Substitution Principle (LSP)
    - Dependency Inversion Principle (DIP)
    - The Interface Segregation Principle (ISP)
  - Cohesion & Coupling
  - Component level design
- User Interface Design
  - Golden Rules for UI Design
  - User Interface Design Process
  - Interface Analysis
  - Design Evaluation Cycle

Unit 3 Quality Assurance

- 15-16. Software Quality
  - What is Quality?
  - McCall's Triangle of Quality
  - The Cost of Quality
  - Achieving Software Quality
  - Software Quality assurance
- 17-19. Testing Strategy & Techniques
  - Testing concepts
  - V&V
  - Testing Strategies
  - Testing techniques
- 20-21. Security Engineering & SCM
  - Why Security Engineering
  - SCM concepts: Baselines / SCI / SCM Process
  - SCM process

Unit 4 Software Project Management

- **22. Project Management Concepts**
  - 4 P
  - W^5HH
- **23. Process and Project Metrics**
  - Why Do We Measure?
  - Process Measurement and Metrics
  - Project Metrics
  - Typical Types of Metrics

Unit 4 Software Project Management

- **24-25. Project Estimation & Scheduling**
  - Scope
  - WBS
  - LOC / FP / COCOMO estimation
  - Scheduling Steps
  - Task Network / Gantt charts / milestone
- **26. Risk Analysis**
  - Project Risk
  - Reactive Risk Management
  - Proactive Risk Management
    - Risk Management Paradigm
    - RMMM